5. Multiplication : Part 1

Revision

Eight dozen bananas were bought to give to the children.

Mahendra : One dozen bananas is 12 bananas. How many are 8 dozen?

Kamal : To find out, we must multiply 12 by 8. This multiplication can be done using the lattice method. So, let's use this method.

12 = 10 + 2

×	10	2
8	80	16

$$\begin{array}{r} 80 \\ +\ 16 \\ \hline 96 \end{array}$$

Eight dozen bananas means 96 bananas.

Kamal : Now answer this. Students are standing in 15 rows for a drill. There are 37 students in each row. How many students are there altogether?

Mahendra : I will tell you. To answer this, I will have to multiply 37 by 15.

×	30	7
10	300	70
5	150	35

37 = 30 + 7 ; 15 = 10 + 5

300 + 150 + 70 + 35 = 555

There are 555 children in all on the field.

Exercise

Multiply the following.

(1) 53 × 8 (2) 25 × 9 (3) 86 × 5 (4) 75 × 11 (5) 41 × 14 (6) 68 × 12

Multiplying a three-digit number

Multiplying whole hundreds by a one-digit number

4 × 100, is also 4 × 1H = 4H = 400; Also, 6 × 100 = 600; 2 × 400 = 800.

◆ **₹ 300 were to be collected from each student for a picnic. 4 students paid on the first day. What was the total amount collected on that day?**

Kamal : To solve this, we will have to take 300, 4 times.

Mahendra : 300 is 3 hundreds. 4 times 3 hundreds, 3H × 4 = 12H, that is 1200.

Kamal : ₹ 1200 were collected on the first day.

Remember : While multiplying whole hundreds by another number, multiply the hundreds digit by the other number and write two zeroes in front of the product.

◆ **If 40 students pay ₹ 300 each, what is the total amount of fees collected ?**

Tony : 40 times 300 = 3 H × 40 = 120 H = 12000. While multiplying 300 by 40, we can multiply the 3 by 4 to get 12; then we can place the 2 zeroes from 300 and one zero from 40 in front of the product 12.

Exercise

Carry out the following multiplications.

(1) 4H × 5 = 2000

(2) 20 × 3H = ☐

(3) 40 × 500 = ☐

(4) 800 × 60 = ☐

(5) 35 × 200 = ☐

(6) ☐ × 70 = 7000

(7) 7 H × 2 = ☐

(8) 20 × 300 = ☐

(9) 600 × 30 = ☐

(10) 900 × 20 = ☐

(11) 600 × 42 = ☐

(12) 15 × ☐ = 30000

Multiplying any three-digit number by a one-digit number

◆ **One set of textbooks costs ₹245. What is the cost of 8 such sets?**

Since the cost of 8 sets of textbooks will be eight times the cost of one set, we will multiply 245 by 8. Remember that 245 = 200 + 40 + 5.

×	200	40	5
8	1600	320	40

$$\begin{array}{r} 1600 \\ +\ 320 \\ +\ 40 \\ \hline 1960 \end{array}$$

Therefore, the cost of 8 sets is ₹1960.

Exercise

1. Multiply.

(1) 124 × 3 (2) 367 × 5 (3) 408 × 9

(4) 627 × 8 (5) 840 × 4 (6) 716 × 7

2. The cost of 1 chair is ₹650. What is the cost of 4 such chairs?

3. One small sack of rice costs ₹825. How much do 5 such sacks cost ?

Multiplying a three-digit number by a two-digit number

◆ **While planting rice, 28 rows have been planted with 244 seedlings in each row. How many seedlings have been planted ?**

There are 244 seedlings in each row. So, there must be 28 times as many seedlings in 28 rows.

We will multiply 244 by 28.

244 = 200 + 40 + 4

28 = 20 + 8

×	200	40	4
20	4000	800	80
8	1600	320	32

	4000
+	1600
+	800
+	320
+	80
+	32
	6832

6832 seedlings have been planted.

◆ **Multiply : 709 × 76**

709 = 700 + 0 + 9

76 = 70 + 6

×	700	0	9
70	49000	0	630
6	4200	0	54

	49000
+	4200
+	630
+	54
	53884

Hence, 709 × 76 = 53884

Exercise

1. Multiply.

   (1) 819 × 12 (2) 545 × 38 (3) 953 × 38

   (4) 610 × 45 (5) 407 × 55 (6) 781 × 90

2. The concessional rate of one English dictionary is ₹175. If 31 students in a class give their teacher that amount for the dictionary, how much money has the teacher collected ?

3. A van carries 205 crates of mangoes. If each crate contains 48 mangoes, how many mangoes are there in the van ?